Composition Pair B1.18 + A1.13 — Content-Domain Specifications Enable Semantic Composition Assessment, and Composition Requirements Provide the Architectural Motivation for Authoring Content-Domain Specifications as Governed Substrate Content

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Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its contribution is to formalize one composition pair: the enabling relationship between content-domain specifications (B1.18) and composition requirements (A1.13), and the motivating relationship that flows in the opposite direction.

Abstract

Within the Coordination Knowledge Substrate (CKS) architecture, entities at every level — cells, aspects, Selves — carry authored content-domain specifications (B1.18): governed substrate content that identifies the operational territory the entity handles. Separately, the architecture specifies composition requirements (A1.13): the conditions an entity must satisfy to be composed into a higher-level entity, assessed through the composition-requirements-five test (A5.14). This note formalizes the enabling pair that links these two concepts. The enabling direction runs from B1.18 to A1.13: content-domain specifications supply the semantic information required for composition compatibility assessment to go beyond technical interface matching. Without B1.18, A1.13 can evaluate only whether entity interfaces are technically compatible; with B1.18, A1.13 can additionally evaluate whether an entity’s operational territory addresses the composing entity’s coordination purpose — a substantively stronger compositional guarantee. The motivating direction runs from A1.13 to B1.18: composition requirements create the governance motivation for authoring content-domain specifications in the first place, since composition assessment requires them. Without A1.13, content-domain might be authored for documentation purposes but would not be architecturally required for the deployment to function. The operational intersection of the pair is B2.92 (content-domain and composition), the Phase B2 note that formalizes how content-domain specifications are applied during composition assessment. Two failure modes are identified: B3.19 (Implicit Content-Domain) when B1.18 is absent; and operational uncoupling

when B1.18 is present but B2.92's application is not activated during A5.14 assessment.

1. Pair identification

B1.18 — Content-domain. Every entity in a CKS architecture — every cell, every aspect, every Self — carries an authored content-domain specification: a statement, expressed as governed substrate content, of the operational territory the entity handles. The content-domain is not a technical interface description; it is a semantic statement of what the entity addresses. A cell's content-domain might specify that the entity handles patient scheduling coordination within a clinical aspect. An aspect's content-domain might specify that the entity coordinates all clinical workflow cells within a particular facility scope. A Self's content-domain might specify that the entity addresses enterprise-wide clinical coordination across multiple facility aspects. Content-domain specifications are authored by governance rather than inferred from runtime behavior; they are substrate content subject to the same human-governed authority, inspect rights, and path retraceability the CKS architecture commits to at every scope (§8.4, §5.2 of the source paper).

A1.13 — Composition requirements. Composition in CKS is governed: a cell does not become part of an aspect, and an aspect does not become part of a Self, through automatic aggregation or runtime discovery. The architecture requires that each entity satisfy explicit composition requirements before being composed into a higher-level entity. These requirements have both a technical dimension — do the entity's interfaces match what the composing entity expects? — and a semantic dimension — does the entity's operational territory address the composing entity's coordination purpose? The composition-requirements-five test (A5.14) is the operational mechanism through which both dimensions are assessed at composition design time.

Relationship type. The pair is an enabling pair with motivating reinforcement: B1.18 enables the semantic dimension of A1.13; A1.13 motivates the authoring of B1.18. Neither concept realizes its full architectural potential alone. B1.18 without A1.13 produces authored specifications that are never used for their primary architectural purpose. A1.13 without B1.18 produces a composition assessment that is technically grounded but semantically unverified. The pair's value is the conjunction.

2. The enabling direction: B1.18 enables semantic A1.13

Without content-domain specifications, composition requirements can only be assessed technically. Technical assessment asks: does this cell's output interface match what the aspect's orchestration rules expect? Does this aspect's interface satisfy the Self's composition requirements at the structural level? Technical compatibility is necessary but not sufficient. A cell with perfectly matching interface types might handle entirely the wrong operational territory for an aspect's coordination purpose. An aspect with compatible interfaces might cover operational territory that overlaps with another aspect already present, creating redundancy, or leave gaps in a Self's intended coverage. Interface compatibility does not catch these problems.

Content-domain specifications per B1.18 supply exactly the semantic information that A1.13's assessment requires. When a governance actor evaluating whether a cell fits within an aspect runs the composition-requirements-five test (A5.14), one of the five dimensions asks: does the cell's operational territory address this aspect's coordination purpose? The answer requires reading the cell's content-domain specification against the aspect's content-domain specification and assessing fit. The same logic applies one level up: does this aspect's operational territory contribute to the Self's integration scope? Again, the answer requires authored content-domain specifications at both levels.

The enabling direction therefore runs from B1.18 to A1.13 through A5.14: content-domain specifications are the substrate content the semantic dimension of the composition-requirements-five test reads. Without B1.18, A5.14 runs in a technically-only mode that cannot assess semantic fit. With B1.18, A5.14 runs in its full mode, assessing both technical and semantic compatibility.

The additional architectural value is not marginal. Semantic composition compatibility assessment prevents two common deployment failures that technical assessment cannot catch. The first is semantic mismatch: a technically compatible entity that addresses the wrong operational domain is composed into an aspect, and the aspect's coordination purpose is not served. The second is content-domain gaps and overlaps: adjacent entities whose content-domain specifications, when compared during composition assessment, reveal either uncovered territory or redundant coverage. Each of these is a first-class conflict per A1.03 requiring governance attention before composition proceeds. Both failures are invisible to technical-only assessment and visible when content-domain specifications per B1.18 are present and applied through B2.92.

3. The motivating direction: A1.13 motivates authoring B1.18

Content-domain specifications require authoring effort. Governance must invest in specifying, for each entity, the operational territory it addresses, expressed as governed substrate content in a form accessible to composition assessment. This investment must be maintained as the entity evolves: DNA evolution per B1.11 may refine an entity's orchestration substrate in ways that shift its effective operational territory, and the authored content-domain specification must be updated to reflect the current scope. Completeness per B2.90's content-domain verification requirements must be satisfied: partial specifications that cover only some of the entity's territory are insufficient for semantic composition assessment.

Without A1.13, the motivation for this investment is weak. Content-domain specifications may be authored for documentation purposes — so human readers understand what each entity handles — or for operational transparency. But documentation can be incomplete, can drift from actual entity behavior, and can lapse when other priorities compete, without breaking the deployment. Documentation-only purposes do not create the structural demand that enforces completeness and currency.

A1.13's composition requirements change the incentive. The architecture requires that composition assessment be performed before entities are composed into higher-level structures. Composition assessment per A5.14 includes a semantic dimension. The semantic dimension requires authored content-domain specifications as its input. Therefore, for any deployment that intends to use composed structures — aspects containing cells, Selves containing aspects — content-domain specifications must be authored, complete, and current. Governance that skips or defers content-domain authoring cannot perform composition assessment with semantic depth.

The motivating direction establishes: A1.13 transforms B1.18 authoring from a useful optional practice into a governance requirement. The presence of A1.13 in the architecture creates a structural demand for B1.18 content that documentation-only purposes cannot create. Governance actors preparing to compose entities are not free to defer content-domain authoring, because the semantic dimension of the required composition assessment depends on it.

4. Mutual dependency

B1.18 depends on A1.13 for architectural application. Content-domain specifications authored per B1.18 serve several purposes: they inform human readers of each entity's scope; they support action-feedback evolution per B1.12 by providing an explicit statement against which scope drift can be detected; they enable directed selection per B1.14 by making scope boundary adjustments explicit; and they contribute to multi-shaped governance per B1.15 by encoding the boundary of each entity's governance jurisdiction. All of these are real architectural values. But the primary architectural application of content-domain specifications — the application that makes their authoring structurally required rather than merely valuable — is in composition assessment per A1.13. B2.92 (content-domain and composition) formalizes this application: it is the operational note that specifies how content-domain specifications are read and evaluated during A5.14. Without A1.13, content-domain specifications remain authored and present in the substrate, but the structural demand for their completeness is absent.

A1.13 depends on B1.18 for semantic depth. Composition requirements assessed without content-domain specifications produce a technically verifiable but semantically shallow guarantee. Interface types can be checked mechanically; structural requirements can be verified against schema. But the architectural claim that a composed deployment is valid — that each aspect's cells address its coordination purpose, that the Self's aspects jointly cover its integration scope, that no semantic gaps or redundancies are present — cannot be substantiated by technical verification alone. The validity claim is stronger, and the deployment is more defensibly governed, when semantic compatibility is verified alongside technical compatibility. Without B1.18, A1.13 produces composition validity claims that are grounded in technical fact but leave the semantic dimension unaddressed.

5. Co-presence requirement

B1.18 without A1.13. Content-domain specifications are authored as governed substrate content, but no composition assessment procedure uses them. The Implicit Content-Domain anti-pattern (B3.19) may develop over time: without the governance pressure that composition assessment creates, content-domain specifications become informal or partial, no longer maintained to the completeness and currency B2.90 requires. The pair's enabling direction is architecturally available but not exercised.

A1.13 without B1.18. The composition-requirements-five test (A5.14) is applied, but the semantic dimension cannot be answered from authored substrate content. Assessment proceeds on technical dimensions only. Composition validity is claimed on a technically-only basis. Content-domain gaps and overlaps are discovered only after deployment when operational behavior reveals mismatches, rather than at composition design time when they can be addressed through directed selection.

Full co-presence — B1.18 and A1.13 with B2.92 coupling. Content-domain specifications are authored per B1.18; composition assessment per A1.13 uses those specifications through B2.92's application in A5.14. The composition-requirements-five test operates on both technical and semantic dimensions. Composition validity claims are both technically and semantically grounded. Gaps and overlaps surface during assessment as first-class conflicts per A1.03, addressable before composition completes.

The critical distinction in the third state is B2.92 coupling. Both B1.18 and A1.13 may be nominally present — content-domain specifications are authored, composition assessment is performed — but the pair can be operationally uncoupled if A5.14 does not read the authored content-domain specifications as input to its semantic dimension. This is the pair's second failure mode, distinct from B3.19, addressed in Section 7.

6. Operational integration at composition design time

The pair's primary operational integration takes place at composition design time, when governance evaluates candidate entities for composition into aspects and aspects for composition into Selves.

The governance actor applying the composition-requirements-five test (A5.14) works through each dimension in sequence. Technical dimensions verify interface compatibility and structural requirements. The semantic dimension — enabled by B1.18 — reads the candidate entity's content-domain specification and the composing entity's content-domain specification to assess operational fit.

B2.92 (content-domain and composition) is the operational note that formalizes how this reading is performed: what specific questions governance asks when comparing content-domain specifications at composition assessment, what constitutes adequate fit, and what the treatment is when fit is partial or absent. For a cell being assessed for aspect membership, the B2.92 check asks: does the cell's content-domain address the aspect's coordination purpose? For an aspect being assessed for Self membership: does the aspect's content-domain address the Self's integration scope?

Two sub-results from B2.92's application are fed back as potential first-class conflicts per A1.03. Content-domain gaps — territory the aspect's coordination purpose requires that no candidate cell's content-domain addresses — are surfaced as governance decisions: is the gap intentional or an omission requiring a missing cell to be identified? Content-domain overlaps — territory that multiple candidate cells address redundantly — are surfaced as potential conflict sources: does the overlap represent deliberate joint coverage or authoring imprecision that should be resolved by refining content-domain boundaries?

Gaps identified at assessment time are addressable through directed selection per B1.14: governance identifies and adds a cell whose content-domain covers the missing territory. Overlaps are resolvable through content-domain boundary refinement: the authored content-domain specifications of overlapping cells are revised to produce non-overlapping coverage under the same human authority that authored them. The composition assessment concludes with composition validity confirmed on both technical and semantic dimensions, or specific gaps or conflicts identified requiring governance action before composition proceeds.

7. Detection and failure modes

Pair detection: four checks.

The B2.92 content-domain and composition check asks: when composition assessment is performed, are content-domain specifications used as input to the semantic dimension of assessment? If content-domain specifications exist as authored substrate content but are not read during A5.14, the pair is technically present but operationally uncoupled.

The A5.14 composition-requirements-five test check asks: does the test include a semantic content-domain compatibility dimension in addition to technical interface checking? A test that proceeds on technical dimensions only indicates either that B1.18 is absent (B3.19) or that B2.92 coupling has not been established.

The B2.93 content-domain verification check asks: do content-domain specifications meet completeness requirements per B2.90? Partial or informal specifications, even when nominally present in the substrate, may indicate B3.19 drift and will produce incomplete semantic assessment even when A5.14 is run with the B2.92 semantic dimension active.

The pair completeness check asks: when A5.14 is run for a given composition decision, does it assess content-domain compatibility in addition to technical interfaces? A yes indicates the pair is coupled. A no indicates uncoupling regardless of whether both B1.18 and A1.13 are nominally present.

Failure mode 1 — Implicit Content-Domain (B3.19). This failure mode develops when B1.18 is absent or allowed to degrade, and A1.13 operates without the semantic input it requires. The symptoms are: composition assessment cannot assess semantic fit; composition validity claims are limited to technical compatibility; content-domain gaps and overlaps are discovered only after deployment when operational behavior reveals mismatches; and there is no authored substrate content against which governance can assess whether an entity's operational territory addresses a composing entity's purpose. The anti-pattern B3.19 identifies the deeper structural problem: content-domain specifications may exist informally in the working knowledge of governance actors, but have not been authored as governed substrate content accessible to composition assessment. The correction is B1.18 authoring performed to the completeness standard B2.90 requires, treating content-domain specifications as first-class substrate content rather than implicit working knowledge.

Failure mode 2 — Operational uncoupling. This failure mode develops when B1.18 and A1.13 are both present, content-domain specifications are authored and current, but the B2.92 application is not established as part of the A5.14 procedure. Content-domain specifications sit in the substrate unused for their primary architectural purpose; composition assessment proceeds on technical dimensions only. The deployment may be technically well-formed and content-domain may be well-authored; but the pair is operationally uncoupled. The correction is establishing B2.92 coupling in the composition assessment procedure: specifying that content-domain specifications are read and their fit assessed as a required step of A5.14.

Asymmetry of the failure modes. B3.19 is the structurally prior failure mode: without B1.18, no coupling is possible regardless of how thoroughly A5.14 is specified. Operational uncoupling is the structurally subsequent failure mode: it requires B1.18 to be present before the uncoupling can be identified as distinct from B3.19. A governance program addressing the pair must confirm B1.18 authoring completeness before checking for operational uncoupling; checking for uncoupling in the presence of incomplete content-domain specifications will produce misleading results.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Related notes in this series

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